

Hospital Chatbot for Patient Color-Coded Clinical Pathways

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Outline of Presentation

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Background: Evolution of Triage

The Concept of Triage

Derived from the French *trier* ("to sort"). Historically evolved from Napoleonic battlefields to modern emergency medicine.

- ▶ **Goal:** Allocate scarce resources based on urgency, not arrival order.
- ▶ **Global Standard:** Time-sensitive conditions (stroke, trauma) require narrow therapeutic windows.
- ▶ **Current Reality (LMICs):** A "bottleneck of access" where patient volume overwhelms clinical staff (?).

Background: The Ghanaian Context

Double Burden of Disease

- ▶ **Infectious:** Malaria, TB, HIV.
- ▶ **NCDs:** Hypertension, Diabetes, Cardiovascular.

Impact on KNUST/KATH

- ▶ Apex referral centers besieged by high volumes.
- ▶ Widespread "No Bed Syndrome" (?).
- ▶ Patients bypass lower-level clinics, causing congestion.

Background: Triage Protocols

- ▶ **Manchester Triage System (MTS):** Europe. Relies on 52 flowcharts and experienced clinical judgment.
- ▶ **South African Triage Scale (SATS):** Africa. Uses Physiology-based scoring (TEWS) + Discriminators. Validated at KATH (?).
- ▶ **Interagency Integrated Triage Tool (IITT):** WHO. Simplified 3-color protocol.

Project Choice

This study adopts a hybrid approach: **SATS logic** within an **Automated Chatbot** using Red/Yellow/Green pathways.

Background: AI and Math Modeling

Mathematical Perspective

An ED functions as a stochastic queuing network.

- ▶ Current state: Arrival (λ) > Service (μ).
- ▶ Result: Exponential growth in waiting times (W_q).

The Shift

Moving from reactive physical triage to **predictive virtual triage**.

- ▶ **NLP:** Extracts symptoms from unstructured text.
- ▶ **Decision Trees:** Maps symptoms to SATS clinical pathways (Explainable AI).

Problem Statement: Systemic Inefficiency

Access Block

- ▶ "No Bed Syndrome" is a failure of information flow, not just physical capacity (?).
- ▶ Uncoordinated self-referral clogs tertiary OPDs with minor (Green) cases.

Resource Constraints

- ▶ Shortage of specialized triage nurses.
- ▶ Manual documentation creates a hard limit on throughput.
- ▶ Wait times often exceed 3 hours (?).

Problem Statement: Mathematical Instability

Queuing Instability

The current system represents an unstable $M/M/s$ queue.

- ▶ As utilization factor $\rho = \frac{\lambda}{s\mu} \rightarrow 1$, the queue length $L_q \rightarrow \infty$.
- ▶ **Current State:** High arrival rate (λ) + Constrained service rate (μ) due to manual vitals/history.
- ▶ **Consequence:** Chaotic waiting rooms and stochastic increase in mortality risk.

The Gap

Existing AI checkers (Babylon/Ada) lack Ghanaian linguistic nuance and specific local clinical pathways (Polyclinic vs. Teaching Hospital).

Objectives of the Study

General Objective

To develop, analyze, and simulate an AI-powered Hospital Chatbot that automates patient triage classification using color-coded clinical pathways to optimize patient flow in Ghanaian hospitals.

Specific Objectives

1. **Develop** an Automated Triage Algorithm based on SATS and Decision Trees.
2. **Guide** Clinical Pathways (Red → Emergency, Green → Self-care) to mitigate congestion.
3. **Optimize** Patient Flow using Queuing Theory (comparing $M/M/s$ vs AI-augmented models).
4. **Integrate** NLP for symptom extraction adapting to local linguistics.
5. **Evaluate** diagnostic accuracy (Sensitivity, Cohen's Kappa) against expert data.

Methodology: Study Design

- ▶ **Approach:** Quantitative, Simulation-based experimental design.
- ▶ **Phases:**
 1. Algorithm Development (Software Engineering).
 2. System Simulation (Mathematical Modeling).
 3. Performance Evaluation (Statistical Analysis).

Rationale

Allows rigorous testing of logic and theoretical impact on hospital operations without clinical risk during development.

Methodology: Mathematical Modeling (Decision Trees)

The chatbot logic uses **Decision Tree Classification** mapping symptoms (Nodes) to Acuity (Leaves).

Splitting Criteria

Uses **Information Gain (IG)** and **Entropy (H)** to determine the most efficient question sequence.

$$H(S) = - \sum_{i=1}^c p_i \log_2 p_i \quad (1)$$

$$IG(S, A) = H(S) - \sum_{v \in \text{Values}(A)} \frac{|S_v|}{|S|} H(S_v) \quad (2)$$

Where S is the dataset and A is the symptom attribute.

Methodology: Queuing Theory Application

To quantify impact, we model the Emergency Department (ED) operations using Kendall's notation.

- ▶ **Baseline Model (Manual):** $M/M/s$ Queue.
- ▶ **Arrivals (M):** Poisson process with rate λ .
- ▶ **Service (M):** Exponential distribution with rate μ .

Key Metric: Average Waiting Time (W_q)

$$W_q = \frac{L_q}{\lambda} = \frac{P_0(\lambda/\mu)^s \rho}{s! \lambda (1 - \rho)^2} \quad (3)$$

Methodology: AI-Augmented Model

The Chatbot acts as an infinite-capacity node ($M/G/\infty$) upstream.

Expected Changes

1. Reduced Effective Arrival (λ'):

$$\lambda' = \lambda(1 - P_{green})$$

2. Increased Service Rate (μ'):

$$\mu' > \mu \quad (\text{due to digital pre-documentation})$$

Methodology: Software Architecture

- ▶ **Backend:** React with Express or Node js (Lightweight/Scalable).
- ▶ **Database:** MySQL (Logs, symptom profiles, audit).
- ▶ **Deployment:** Vercel for Frontend and Render for backend

Clinical Logic Source

Strictly populated using **South African Triage Scale (SATS)** discriminators (e.g., Emergency Signs, TEWS logic).

Methodology: NLP Pipeline

Handling unstructured inputs (e.g., "I have a running belly").

1. **Tokenization:** Breaking text into units.
2. **Stemming:** Lancaster Stemmer (e.g., "bleeding" → "bleed").
3. **Vectorization:** Bag-of-Words (BoW) model.
4. **Intent Classification:** Deep Neural Network (DNN) with ReLU activation and Softmax output.

Methodology: Simulation Data

Data Source

Synthetic patient arrival data generated based on statistical distributions (Poisson) observed in Ghanaian hospital studies.

- ▶ **Reference Data:** AngloGold Ashanti Hospital study (?).
- ▶ **Parameters:** $\lambda \approx 53$ patients/hr; $s = 5$ doctors.

Methodology: Ethics and Assumptions

- ▶ **Human-in-the-Loop:** Chatbot is a Decision Support System (DSS), not a replacement. "Red" cases trigger immediate hospital attendance.
- ▶ **Privacy:** Anonymized logs, compliant with Ghana Data Protection Act.
- ▶ **Assumption:** Patients possess basic literacy/access to smartphone or kiosk.

Results: Triage Accuracy Metrics

Evaluated against "Gold Standard" historical decisions.

Sensitivity (Recall)

$$\text{Sensitivity} = \frac{\text{TruePositives}_{Red}}{\text{TruePositives}_{Red} + \text{FalseNegatives}_{Red}}$$

Targeting **100% sensitivity** for Critical cases (Fail-safe principle).

Inter-Rater Reliability

Cohen's Kappa (κ): Recent studies suggest $\kappa \approx 0.695$ (Substantial Agreement) for AI triage (?).

Results: Baseline Queue Simulation

Scenario: Manual Triage at AngloGold Ashanti parameters.

- ▶ $\lambda = 53$ patients/hr.
- ▶ Effective Service Capacity (5 doctors).
- ▶ Utilization $\rho \approx 0.78$.

Outcome

Calculated Average Waiting Time (W_q): **2.41 hours**.
aligns with empirical observations of congestion (?).

Results: AI-Augmented Scenario A

Mechanism: Service Time Reduction via pre-documentation.

- ▶ Chatbot pre-collects history/demographics.
- ▶ Physical nurse service time ($1/\mu$) reduces by $\sim 20\%$.

Result

Non-linear drop in waiting time. Pushes system away from the "knee" of the exponential curve.

Potential reduction: $> 50\%$.

Results: AI-Augmented Scenario B

Mechanism: Load Diversion (Load Balancing).

- ▶ Diversion of "Green" (non-urgent) patients to Polyclinics.
- ▶ Effective arrival rate drops: $53/hr \rightarrow \sim 37/hr$ (assuming 30% diversion).

Result

Server utilization drops significantly.

Waiting time approaches negligible levels (minutes).

Results: Clinical Pathway Mapping

Standardization of the triage process ("Glass Box" Logic).

Symptom Input	SATS Discriminator	Output / Action
"Can't breathe, chest hurts"	Difficulty Breathing	RED: Immediate ER
"Stomach pain, very bad"	Severe Pain	YELLOW: Urgent Care (<60m)
"Runny nose, cough"	None (Routine)	GREEN: Polyclinic/Pharmacy

Table: Chatbot Pathway Logic Mapping

Traceable logic ensures accountability, unlike "Black Box" Neural Networks.

Results: Operational Efficiency

- ▶ **Baseline:** Staff utilization $> 88\%$ (Burnout zone).
- ▶ **AI Model:** Utilization drops to manageable 60 – 70%.

Surge Capacity

This "slack" is vital for safety, creating a buffer for mass casualty events. The chatbot acts as a digital **load balancer**.

Deep Analysis: The "Glass Box" Advantage

- ▶ **Constraint Satisfaction:** SATS uses rigid rules (e.g., "If oxygen $< 90\%$ $\rightarrow$ RED").
- ▶ **Safety:** Rigid rules are safer for triage than probabilistic guesses used in pure Deep Learning.
- ▶ **Hybrid:** NLP (Probabilistic) for understanding $\rightarrow$ Decision Tree (Deterministic) for safety.

Deep Analysis: Cultural Linguistics

The Semantic Gap

"Running belly" (Diarrhea) or "Waist pain" (Lower back pain).

- ▶ Western-trained models fail on these idioms.
- ▶ **Solution:** Training on localized corpora (Masakhane Project datasets) is a safety requirement, not just a feature (?).

Conclusion

Scientific Validity

Integrating SATS protocols with Decision Tree algorithms creates a robust tool mirroring expert logic.

Operational Impact

Mathematical modeling proves that reducing effective arrival rate (λ) and increasing service rate (μ) addresses the root causes of overcrowding.

Addressing "No Bed Syndrome"

Alleviates the syndrome by preventing misuse of tertiary resources and guiding non-urgent patients away from the emergency center.

Recommendations: Implementation

1. **Pilot Implementation:** Deploy via kiosks at KNUST Hospital or KATH OPD entrances to collect real-world validation data.
2. **Voice Interaction:** Integrate ASR for local languages (Twi, Ga, Hausa) to ensure equity.
3. **Human-in-the-Loop:** Maintain "triage verification" by a nurse. The bot is a handover tool, not a doctor.

Recommendations: Systemic Integration

- ▶ **National Strategy:** Evaluate for integration with Ghana Health Service (GHS) database for real-time bed availability checks.
- ▶ **Green Lanes:** Hospitals must establish "Fast Track" clinics for Green-triaged patients so diversion does not equal denial of care.

Thank You